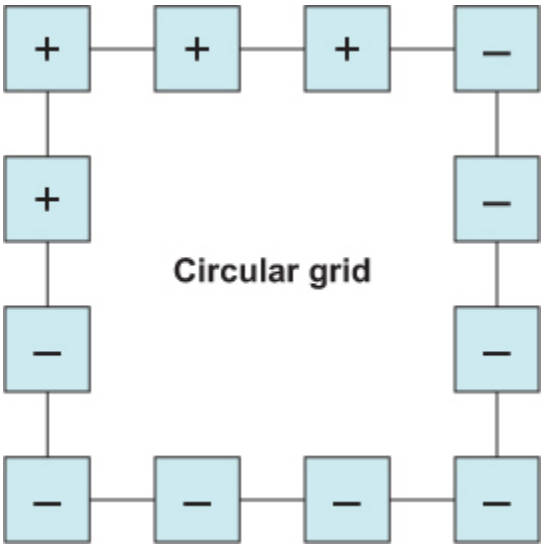
- ***9*** We take the action in our temporary copy of the grid, grid_, and only once all agents have taken actions do we copy them into the main grid.
- ***10*** The target value is the Q value vector with the Q value associated with the action taken replaced with the reward observed.
- ***11*** Manual gradient descent
- ***12*** Copies the contents of the temporary grid_ into the main grid vector.

In the main training loop, we iterate through all 20 agents (which are representing electrons), and for each one we find its left and right neighbors, get their joint-action vector, and use that to compute Q values for the two possible actions of spin-down and spin-up. The 1D Ising model, as we've set it up, isn't just a line of grid cells but rather a circular chain of grid cells such that all agents have a left and right neighbor (figure 9.15).

Figure 9.15. We are representing the 1D Ising model with a single binary vector, but it is actually a circular grid because we treat the leftmost electron as being immediately next to the rightmost electron.



Each agent has its own associated parameter vector that we use to parameterize the Q function, so each agent is controlled by a separate deep Q-network (although it is only a 2-layer neural network, so not really "deep"). Again, since each agent has the same optimal policy, which is to align the same way as its neighbors, we could have used a single DQN to control them all. We will use this approach in our subsequent projects, but